FORMAL SMOOTHNESS OF THE SOURCE MAKES THE CONORMAL-AGAINST-DIFFERENTIALS SEQUENCE SHORT EXACT ON SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4985-4998 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5000-5013, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). Paired with same-wave L16226.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas universally-unramified-differentials-sequence (L16226), differentials-relative-immersion, differentials-relative-immersion-section; universal first order thickening existence; formal smoothness lifting over split first order thickenings.

Source reference. Original lemma 05ZB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1180–1191. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04FC: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3366–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let $h : Z \to X$ be a formally unramified morphism of algebraic spaces over B. Assume that Z is formally smooth over B. Then the canonical exact sequence*

$$0 \to \mathcal{C}_{Z/X} \to h^*\Omega_{X/B} \to \Omega_{Z/B} \to 0$$

of Lemma 04FC is short exact.

Proof. Let $Z \to Z'$ be the universal first order thickening of Z over X. From the proof of Lemma 04FC we see that our sequence is identified with the sequence

$$\mathcal{C}_{Z/Z'} \to \Omega_{Z'/B} \otimes \mathcal{O}_Z \to \Omega_{Z/B} \to 0.$$

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Since $Z \to S$ is formally smooth we can étale locally on Z' find a left inverse $Z' \to Z$ over B to the inclusion map $Z \to Z'$. Thus the sequence is étale locally split, see Lemma 05ZB. $\square$